

Design Your Own LOW COST RISC-V SYSTEM

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Traditionally, the integration of microcontrollers into electronic devices has been a substantial contributor to the bill of materials (BoM) and, consequently, the overall cost of the product. Devices ranging from game pads and electronic controllers to sophisticated robotics often demand microcontrollers with low prices. This financial barrier has limited the accessibility of advanced technology to a broader audience.

This system aims to reduce costs associated with the microcontroller. The integration of RISC-V architecture provides an efficient and cost-effective solution for designing such devices as induction heaters, while maintaining affordability (under hundred rupees). The components

used in the design are listed in the Bill of Materials table.

The CH32V003F4U6 RISC-V microcontroller is a game-changer in cost-effective system design. It not only significantly reduces the cost associated with the heart of the system but also paves the way for innovative solutions where cost efficiency is paramount. Fig. 1 shows the RISC-V chip CH32V003.

Power system design

According to the datasheet of the chip, it works within the voltage range of 5 to 3.3V. So the voltage regulation system using LM1117 with two capacitors has been designed to provide regulated power supply of 3.3V.

The voltage regulation circuit consists of the LM1117 voltage regulator, two capacitors, and the CH32V003F4U6 microcontroller. The LM1117 is connected with its input pin to the unregulated voltage source and its output pin to the microcontroller, providing a stable 3.3V output. Fig. 2 shows the power system circuit and the SMD soldering of components.

BILL OF MATERIALS

Name	Designator	Quantity
22pF capacitor	C1, C2	2
100nF capacitor	C3, C6	2
10 μ F, 16V capacitor	C4, C5	2
LED0805	LED1, LED2	2
1k Ω resistor	R1, R2	2
CH32V003F4U6 RISC-V chip	U1	1
XXHCLNANF-24MHz crystal	U2	1
GT-TC054C-H035-L1W push button	U3	1
USB-MICRO_B connector	U4	1
PM127V-11-12-H46 header pin	U5, U7	2
LM1117GSF-3.3 voltage regulator	U6	1

Capacitor configuration

Two capacitors are strategically connected across the input and output pins of the LM1117 to enhance stability and filter out noise:

Input capacitor (C_{IN}). Connected between the input pin of LM1117 and the ground, helps stabilise the voltage input and mitigate transient fluctuations.

Output capacitor (C_{OUT}). Connected between the output pin

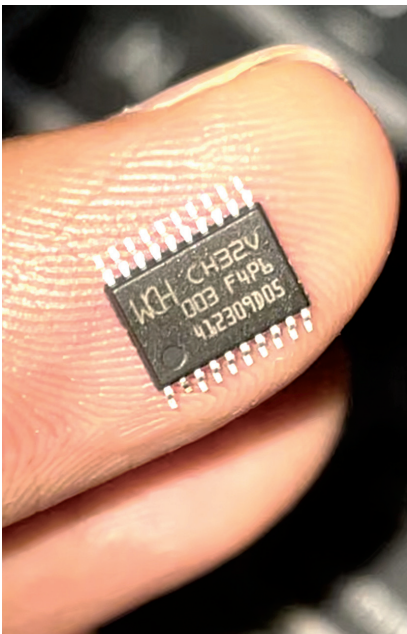


Fig. 1: RISC-V chip CH32V003

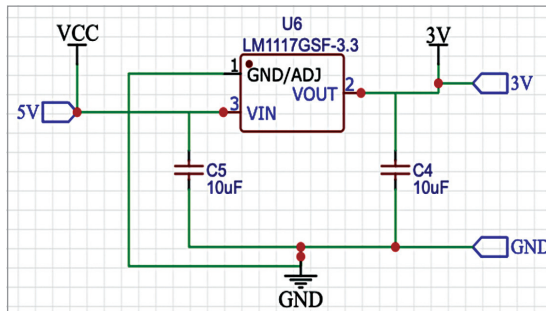


Fig. 2: Power system circuit and SMD soldering of components

